

Patrick (Yong Jae) Kwon

Mail : yongjae.kwon@ucf.edu

Linkedin : <https://www.linkedin.com/in/patrickkwon>

Github : <https://www.github.com/yj7082126>

RESEARCH INTERESTS

My main research goal is in embedding human-level spatial intelligence towards generative models, in order to provide better user controllability in solving various problems.

EDUCATION

University of Central Florida Computer Science Ph.D in Computer Science , Advisor: Dr. Chen Chen	08/25 - 05/28 (Anticipated) Orlando, FL
Columbia University Data Science Master of Science (GPA : 3.9/4.0)	08/2018 – 12/19 New York, NY
University of Virginia Computer Science, Statistics Bachelor of Arts (GPA : 3.8/4.0) (2 year early graduation)	08/15 – 05/17 Charlottesville, VA

WORK EXPERIENCE

Camera and Computer Vision Development Intern ByteDance Inc	05/26 - 08/26 San Jose, CA
Deep Learning Researcher Naver Webtoon	09/21 - 08/25 Pangyo, ROK
Deep Learning Researcher Deepbrain AI	01/20 - 09/21 Seoul, ROK
Research Assistant Columbia University CGUI Lab	09/19 – 12/19 New York, NY
Data Scientist Emadri	06/19 – 12/19 New York, NY
Data Analyst Krafton	06/17 – 07/18 Pangyo, ROK

PUBLICATIONS

* indicates equal contribution

1. C. Banarjee, A. Bilic, P. Kwon, S. Rojas Aristizabal, J. Lopez, R. Xie, C. Chen, and L. Thiamwong. Use of computer vision with conventional video recordings of gait in older adults: A scoping review. *Healthcare*, 14(16), 2026. doi: 10.3390/healthcare14162487
2. P. Kwon and C. Chen. Factorizedhmr: A hybrid framework for video human mesh recovery. In *Advances in Neural Information Processing Systems*, May 2026. doi: 10.48550/arXiv.2605.14854
3. C. Banarjee, P. Kwon, A. Lipat, R. Xie, C. Chen, and L. Thiamwong. Fall risk and gait analysis in community-dwelling older adults using world-spaced 3d human mesh recovery. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, April 2026. doi: 10.48550/arXiv.2604.11961
4. P. Kwon and C. Chen. Dreamingcomics: A story visualization pipeline via subject and layout customized generation using video models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2026. doi: 10.48550/arXiv.2512.01686

5. P. Kwon, C. Chen, and H. Joo. Graspdiffusion: Synthesizing realistic whole-body hand-object interaction. *The IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, March 2026. doi: 10.48550/arXiv.2410.13911
6. H. Kim, S. Han, P. Kwon, and H. Joo. Beyond the contact: Discovering comprehensive affordance for 3d objects from pre-trained 2d diffusion models. *European Conference on Computer Vision*, September 2024. doi: 10.1007/978-3-031-72983-6_23
7. S. J. Song, M. Tang, B. Gwartzman, D. Lee, P. Romandini, M. Salem, P. Kwon, S. K. Feiner, and I. Sailer. Augmented-reality-assisted intraoral scanning: A proof-of-concept study. *Journal of Prosthodontics*, 33(6):550–557, July 2024. doi: 10.1111/jopr.13836
8. B. Kim*, P. Kwon*, K. Lee, M. Lee, S. Han, D. Kim, and H. Joo. Chupa: Carving 3d clothed humans from skinned shape priors using 2d diffusion probabilistic models. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 15919–15930, October 2023. doi: 10.1109/ICCV51070.2023.01463
9. K. Lee*, P. Kwon*, M. Lee, N. Ahn, and J. Lee. LPMM: Intuitive pose control for neural talking-head model via landmark-parameter morphable model. *arXiv preprint arXiv:2305.10456*, May 2023. doi: 10.48550/arXiv.2305.10456
10. N. Ahn, P. Kwon, J. Back, K. Hong, and S. Kim. Interactive cartoonization with controllable perceptual factors. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16827–16835, June 2023. doi: 10.1109/CVPR52729.2023.01614
11. P. Kwon, J. You, G. Nam, S. Park, and G. Chae. Kodf: A large-scale korean deepfake detection dataset. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10724–10733, 2021. doi: 10.1109/ICCV48922.2021.01057

PROJECTS AND RESEARCH

Seeing your way Indoors Python	07/26
ByteDance PICO	
Created a indoor navigational system based on scene reconstruction and localization pipelines.	
AI Photocard Python	12/23
Naver Webtoon AI	
Built a python-based library to service the state-of-the-art diffusion models in generating image, video and 3D content, for internal usage	
Augmented-Reality-Assisted Intraoral Scanning (ARIOS) C#	06/23
Columbia University	
Participated in a proof-of-concept study of implementing Augmented Reality towards intraoral scanning to further improve efficiency of scanning procedures.	
Pally : Augmented Reality for Social Transition C#	06/19
Verizon 5G Edtech Challenge	
Project on improving social skills for autistic children using Microsoft Hololens and 5G Network.	

HONORS AND AWARDS

Verizon 5G Edtech Challenge Winning Project	05/19
for "Pally: Augmented Reality for Social Transition" (Top 10 amongst submitted projects)	
UVA Order of the Orange Stole	05/17
Recognition for early graduation at University of Virginia	
Dean's List	08/15 – 05/17
Recognition for academic excellence at University of Virginia	